AI Gun



Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: AI gun

What it is used for

An artificial insemination (AI) gun is used by trained animal-reproduction personnel to deposit prepared semen at the appropriate site during an approved breeding procedure. Correct hygiene, temperature control, and gentle handling help protect semen quality and animal welfare.

Main parts

- Outer barrel holds the prepared insemination sheath or compatible assembly.
- Plunger or push rod advances the semen dose.
- Handle provides a grip for controlled insertion and delivery.
- Locking or retaining mechanism secures compatible sheaths where fitted.
- Protective cover maintains cleanliness before use.
- Compatible insemination sheath guides the instrument during the procedure.
- Identification label or marking helps confirm the intended semen dose.
- Cleaning and storage accessories support hygienic reuse where permitted.

Before you start

1. Only trained and authorised animal-reproduction personnel should perform insemination.
2. Confirm the animal's identity, breeding record, and suitability for the planned procedure.
3. Check semen identification, expiry or approved use period, and handling requirements.
4. Inspect the AI gun and compatible sheath for damage, cleanliness, and correct fit.
5. Prepare clean gloves, lubricant, and other supplies required by the approved procedure.
6. Keep frozen semen in the approved storage system until it is needed and follow the specified thawing protocol.

How to use

1. Wash hands and put on the required clean protective gloves.
2. Verify the semen identification against the animal's breeding record before preparation.
3. Prepare and load the semen straw and sheath according to the equipment and semen instructions.
4. Secure the sheath and confirm the plunger moves correctly without forcing it.
5. Prepare the animal using the approved hygiene and restraint procedure.
6. A trained operator should position and guide the instrument gently using the species-specific approved technique.
7. Deposit the semen at the appropriate site according to the established breeding protocol.

8. Withdraw the instrument carefully and dispose of single-use components in the designated waste stream.
9. Clean and disinfect reusable components using the approved method, if the device is designed for reuse.
10. Record the semen identification, animal details, date, operator, and relevant procedure observations.

⚠ Safety

- Never perform insemination without appropriate training, authorisation, and species-specific procedure knowledge.
- Use clean, compatible equipment and avoid contaminating the semen or sheath.
- Do not force the instrument or continue if resistance, pain, bleeding, or injury occurs.
- Handle animals with suitable restraint and assistance to prevent injury to the animal and operator.
- Follow approved semen storage and thawing instructions; avoid uncontrolled temperature exposure.
- Dispose of contaminated and single-use materials according to veterinary hygiene and waste procedures.

Care & storage

- Clean and disinfect reusable components immediately after use according to the manufacturer instructions.
- Inspect the barrel, plunger, and retaining parts for wear or damage.
- Store the dry, clean instrument in a protective case away from dust and chemicals.
- Maintain breeding and equipment records according to the facility procedure.

Common mistakes to avoid

- Using semen without verifying its identification and handling requirements.
- Touching or contaminating the prepared sheath or semen dose.
- Forcing the instrument when insertion meets resistance.
- Failing to record the animal, semen dose, and insemination details.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.